

Step-by-Step Guide for Creating an S3 Bucket and Uploading a file

We will be creating an Amazon S3 bucket to support our project data. The steps to create the bucket are outlined below, each accompanied by a brief explanation to clarify its importance and relevance to our use case.

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Step 1 - Sign in to AWS and Navigate to S3.

- **Sign in** to the **AWS Management Console**, locate “**Services**” in the top menu (left side), and select S3 under “Storage” or directly search S3 bucket as shown in figure 1.

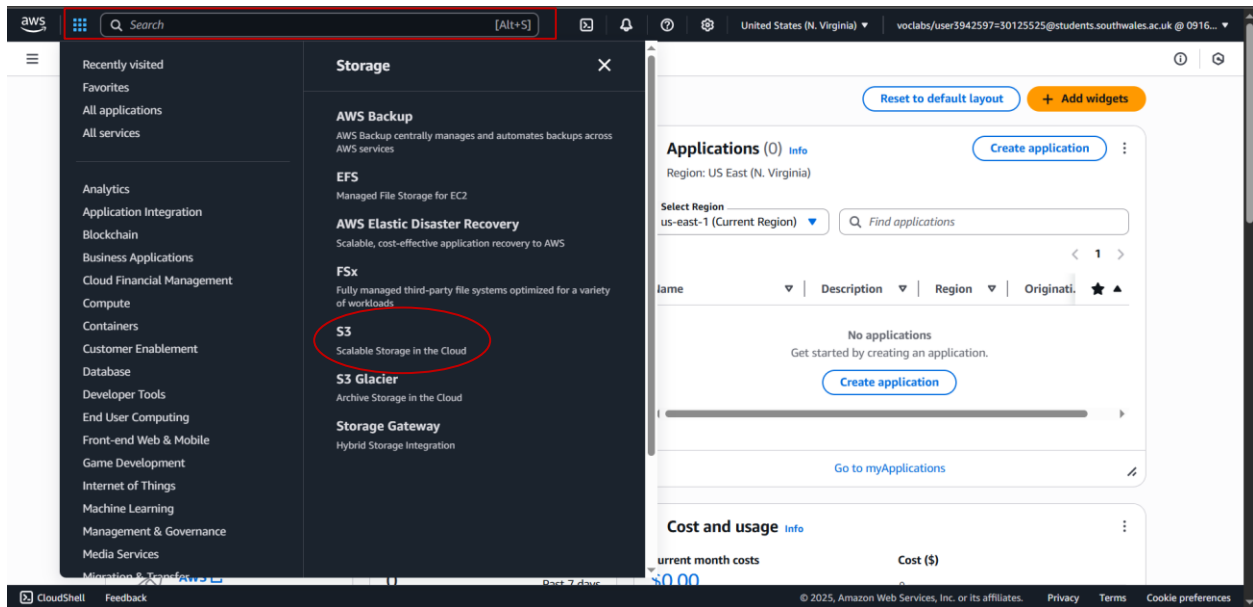


Figure 1. Navigating to S3.

Step 2 - Create a New Bucket.

- To create a new bucket, click **“Create bucket”** as marked in figure 2.

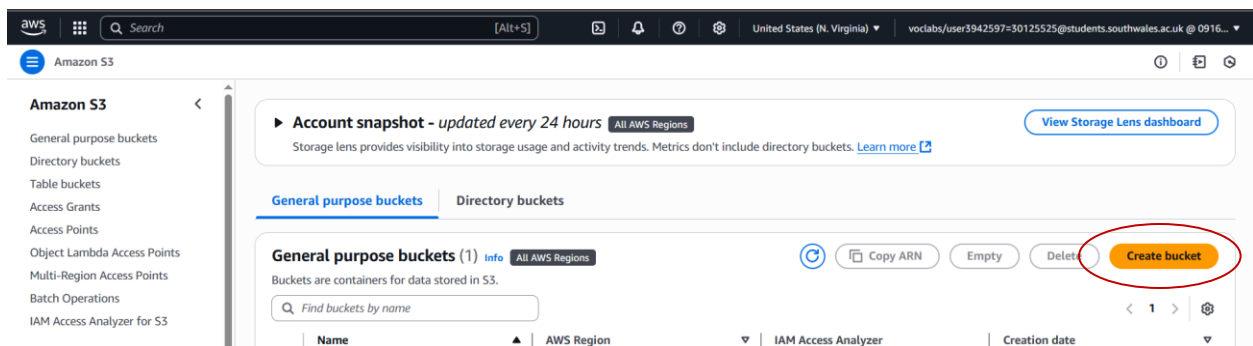


Figure 2. S3 Dashboard.

- In the **“Bucket name”** field enter **30125525-data-safe-insurance-bucket**. We need to have a unique name to recognize and navigate our data stored in that bucket. In Figure 3, our bucket name is **“30125525-data-safe-insurance-bucket.”**

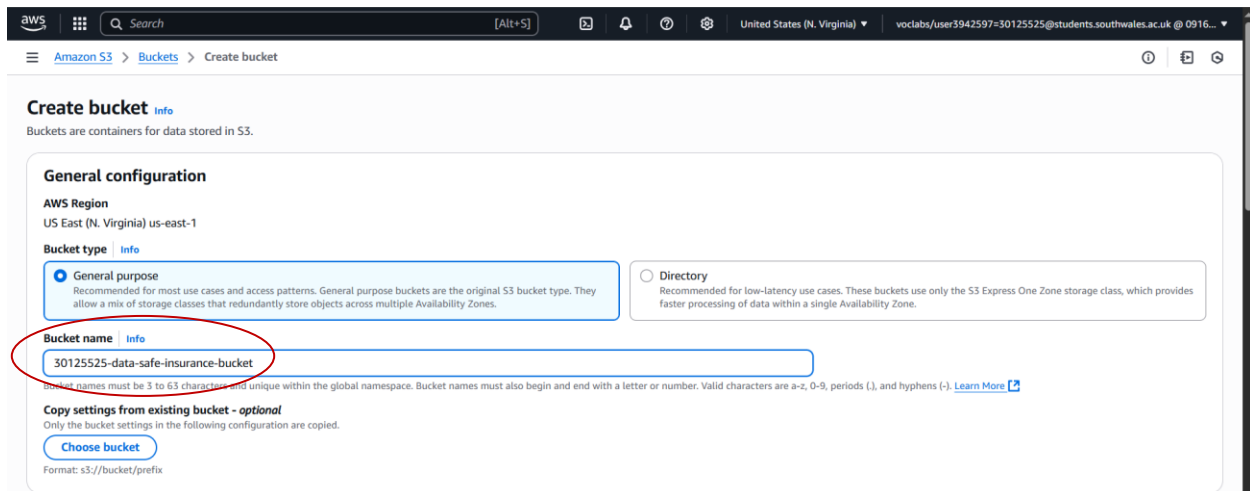


Figure 3. General Configuration, assigning a bucket name.

Step 3 - Select AWS Region.

- Here, we are using “**US East (N. Virginia) us-east-1**” since we do not have permission to create S3 bucket in another region.

If we choose a closer region it reduces latency and may lower transfer costs. It also helps meet any data-residency requirements. If we must change region do as shown in figure 4.

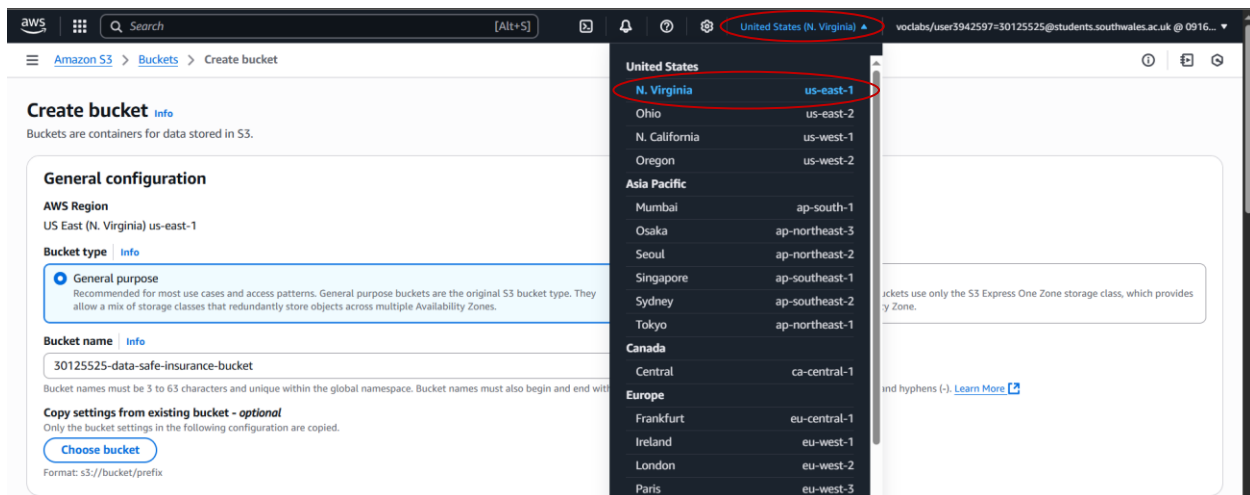


Figure 4. Selecting a region while creating a bucket.

Step 4 - Block Public Access.

- Keep “**Block all public access**” **enabled** (all four checkboxes ticked) as shown in figure 5.

By default, S3 buckets are private. Blocking public access ensures no objects are exposed unintentionally.

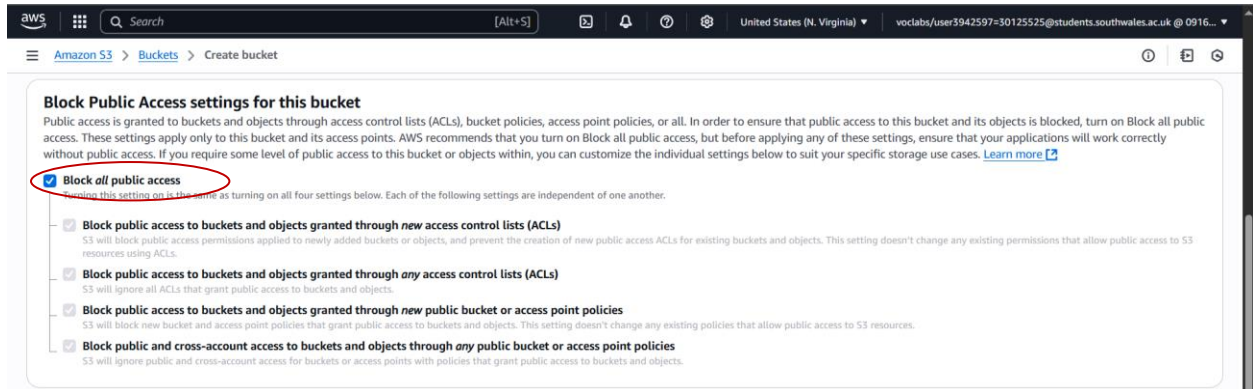


Figure 5. Block Public Access settings for S3 Bucket.

Step 5 - Bucket Versioning.

- Under “**Bucket Versioning**,” choose **Enable**.

Versioning protects against accidental overwrites or deletions by keeping prior versions of every object with timestamps.

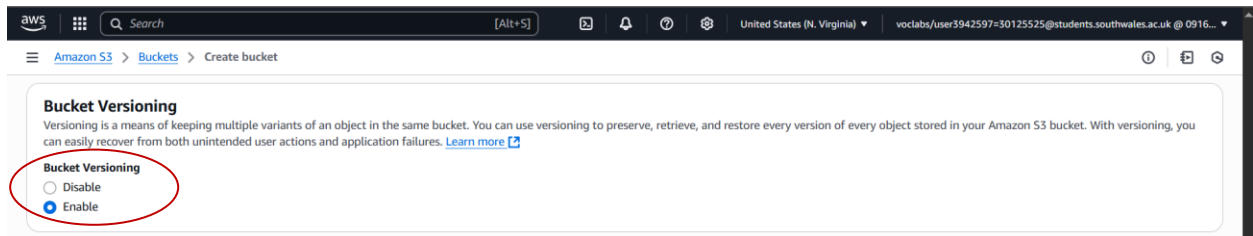


Figure 6. Enabling Bucket Versioning.

Step 6 – Name Tags.

- Under “**Tags**,” add Key/Value pairs such as “**Project: Insurance, Owner: DataSafe Insurance**” as shown in figure 7.

Tags enable cost allocation, lifecycle automation, and resource identification in large environments.

Tags - optional (2)
You can use bucket tags to track storage costs and organize buckets. [Learn more](#)

Key	Value - optional	
Project	Insurance	Remove
Owner	DataSafe Insurance	Remove

[Add tag](#)

Figure 7. Tagging S3 bucket under a specific Project.

Step 7 - Default Encryption.

- Initially the first option is selected by default, and we will keep it as it is that is “SSE-S3” as shown in figure 8.

Encrypting data at rest is a security best practice and often a compliance requirement.

Default encryption [Info](#)
Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)

- ☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)
- ☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
- ☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Secure your objects with two separate layers of encryption. For details on pricing, see [DSSE-KMS pricing](#) on the [Storage](#) tab of the [Amazon S3 pricing page](#).

Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

- ☐ Disable
- ☒ Enable

Figure 8. Encryption for S3 Bucket.

Step 8 - Create Bucket.

- Final step is to review all our configurations and the finally click on the **bottom right** button “**Create Bucket**” as shown below in figure 9.
- After creating a bucket, we can find it in **General Purpose buckets** on the left panel as shown in figure 10.
- Next step is to upload the Insurance Data provided.

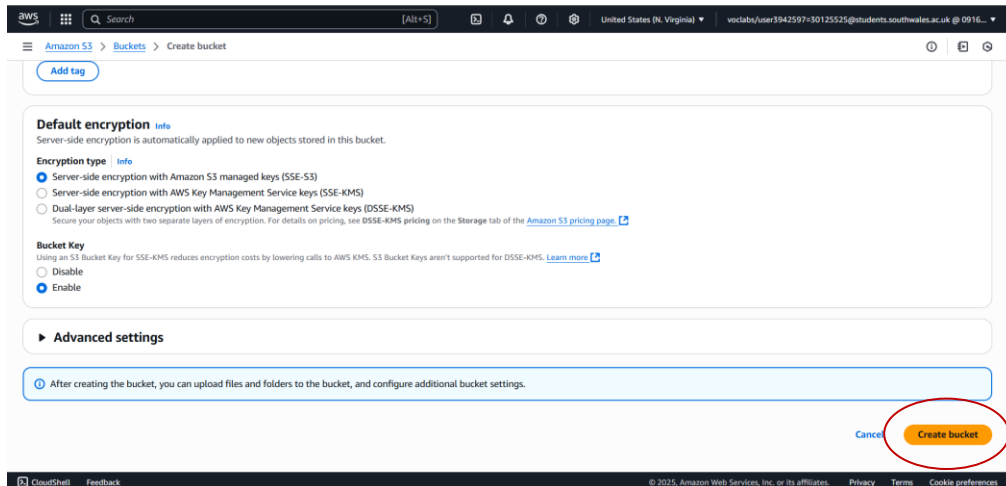


Figure 9. Final step to create a bucket.

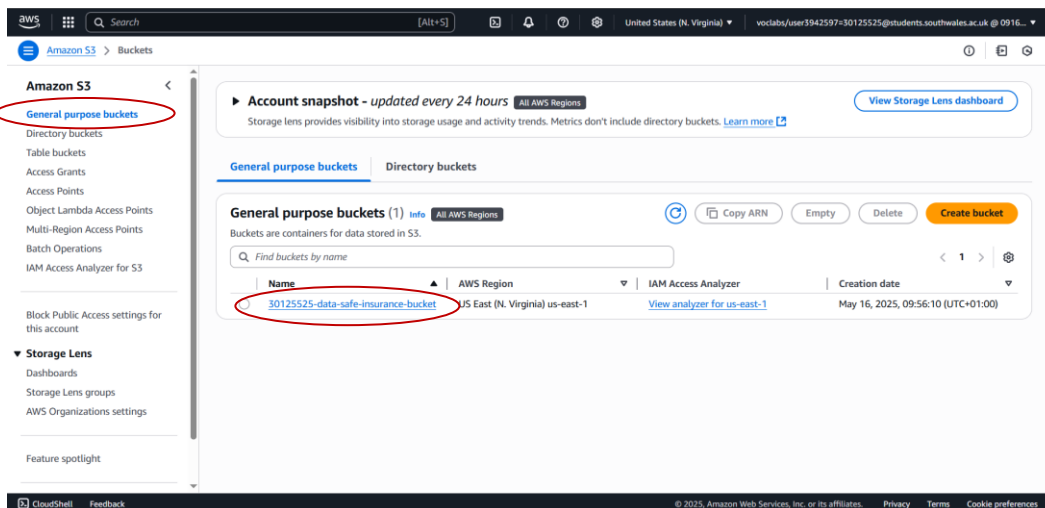


Figure 10. Bucket created after final configurations.

Step 9 - Uploading Insurance Data.

- After clicking the bucket, we will be uploading a data file. Here, we have a “JSON” file.
- Click the “Upload” button as shown in figure 11.

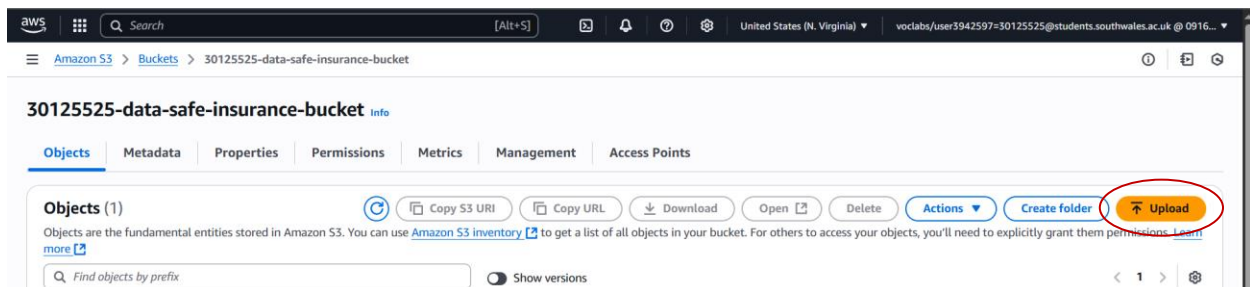


Figure 11. Uploading JSON file.

Step 10 - Drag and drop the desired file.

- AWS provides an option to **drag and drop** the file we want in our Bucket. Here, we have uploaded our JSON file called “**insurance-dataset-2024-2025-dbjson.json**”.

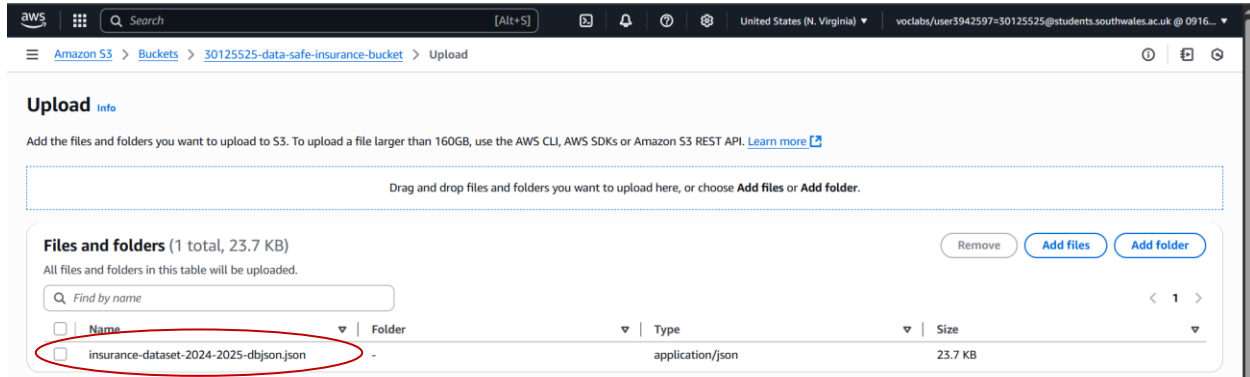


Figure 12. Drag and Drop JSON file.

Step 11 - Picking Storage Class.

- We must pick a suitable Storage class based on our data.
- In Figure 13, **Standard class** seems a good fit for our type of data since we will be frequently accessing it.

Other classes have minimum object size and duration. We picked Standard to keep it flexible.

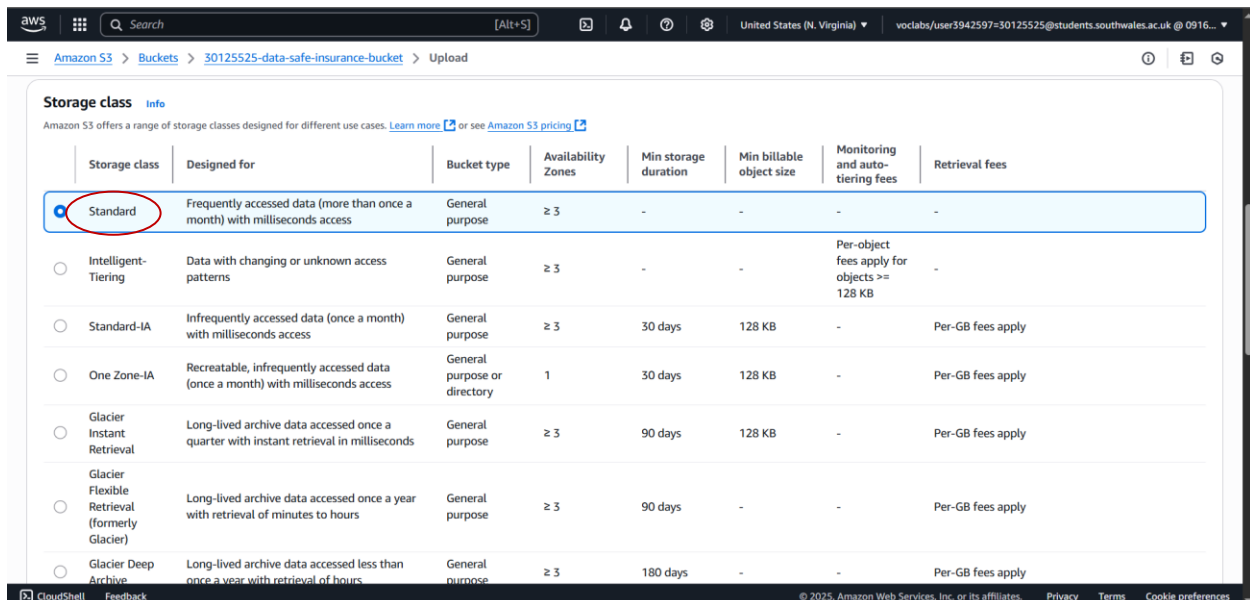
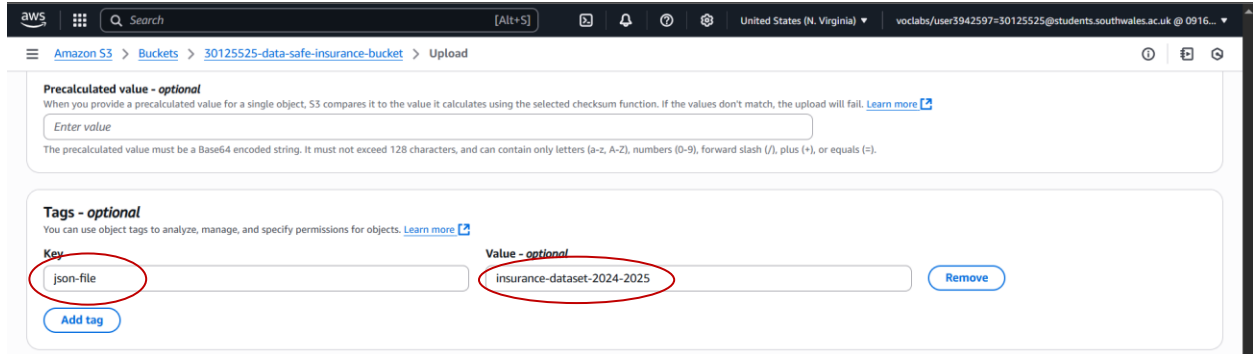


Figure 13. Picking a suitable Storage Class.

Step 12 – Tags.

- Under “Tags,” add Key/Value pairs such as “**json-file: insurance-dataset-2024-2025**” as shown in figure 14.

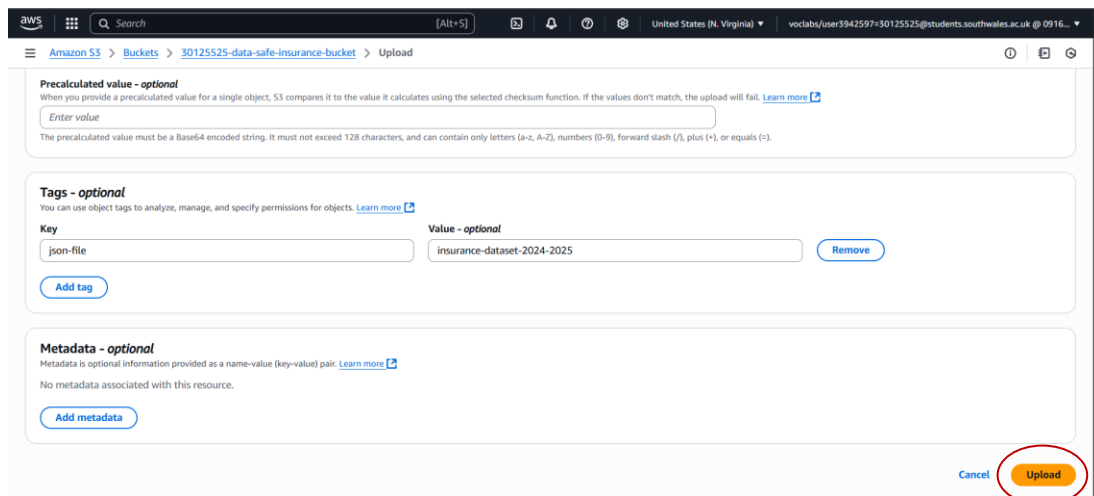


The screenshot shows the AWS S3 console's 'Upload' page for a bucket named '30125525-data-safe-insurance-bucket'. The 'Tags - optional' section is highlighted with a red circle. It contains a table with two columns: 'Key' and 'Value - optional'. The 'Key' column has the value 'json-file' and the 'Value - optional' column has the value 'insurance-dataset-2024-2025'. Both fields are circled in red. There is an 'Add tag' button below the 'Key' field and a 'Remove' button next to the 'Value' field. Above the tags section, there is a 'Precalculated value - optional' section with an 'Enter value' input field and a 'Learn more' link.

Figure 14. Adding tags for the Insurance Data.

Step 13 – Final step to upload JSON file.

- After reviewing all the step click the “**Upload**” button at the bottom right of the page.



The screenshot shows the same AWS S3 console 'Upload' page. The 'Tags - optional' section is still visible. Below it, the 'Metadata - optional' section is shown, with a note that no metadata is associated with this resource. At the bottom right of the page, the 'Upload' button is circled in red, indicating the final step to complete the upload process. The 'Cancel' button is also visible next to it.

Figure 15. Final step to upload a file.

- After all the steps have been performed we can finally see that our “JSON” file is uploaded in our specified bucket in the name of “**insurance-dataset-2024-2025-dbjson.json**” as shown in the figure 16 below.

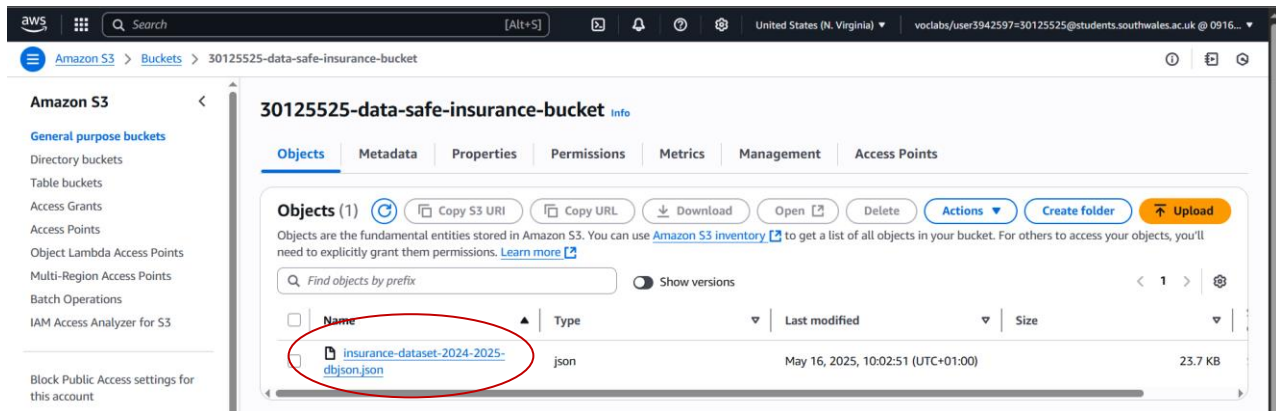


Figure 16. Final snippet after uploading the image.

References

[1] Amazon Web Services, “What is Amazon S3?,” Amazon Simple Storage Service Documentation, 2024. [Online]. Available: <https://docs.aws.amazon.com/AmazonS3/latest/userguide/Welcome.html>. [Accessed: 18 May 2025].

[2] Amazon Web Services, “Getting started with Amazon S3,” Amazon Simple Storage Service Documentation, 2024. [Online]. Available: <https://docs.aws.amazon.com/AmazonS3/latest/userguide/GetStarted.html>. [Accessed: 20 May 2025].

[3] Amazon Web Services, “Understanding and managing Amazon S3 storage classes,” Amazon Simple Storage Service Documentation, 2024. [Online]. Available: <https://docs.aws.amazon.com/AmazonS3/latest/userguide/storage-classes.html>. [Accessed: 20 May 2025].